AP157 Output 9 - April 10, 2026

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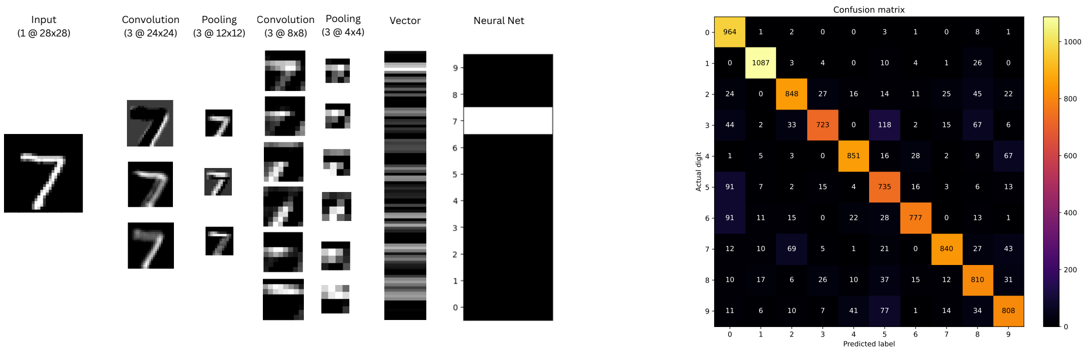

Figure 1: **CNN Workhorse** A figure showing how the input is passed into convolutions to get filters/edges independently and pooled to limit volatility in training, and read what the input is. The right plot shows the matrix of the frequency of how a certain input yileds a particular output

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CNN Workhorse

From NIST, we have inputted an OCR data of size 28×28. This is going to be convolved and pooled twice, alternately. The convolution is described by the equation:

$$C = I - K + 1 \tag{1}$$

where I is the dimensions of the input (in this case, 28), K is our kernel size (which we will choose to be 5 ; this means the kernel is 5×5). This means that the first convolution output would have three feature maps of size 24×24.

What do we choose as kernels here? We iterate $K \times K$ submatrix overlappingly across the entire matrix and multiply each to the "mother" matrix to get features. With this, the image is learned without doing the Sobel Operator which is only useful for idnetifying edges, not finding features. Convolutions learn it by multiplying with its associated kernel, and their interactions with another kernels can turn them into "edge" or not. Choosing K to be too small leaves the model unable to study the variations much like a smalling rate in gradient descent, and a K too large makes the variations coarser than the signal.

Then Pooling reduces the output of convolution by a factor P, where P is our pooling dimensionality. If convolution activates, this averages to act against noise sensitivity ensuring robust understanding even when the validation dataset deviates decently from the training dataset. They do not overlap and as such,

$$O = \frac{C}{P} \tag{2}$$

We choose P to be 2 as an excessively large value would result in a blurred feature map.

As such, the first pooling would keep the number of feature maps at three but reduces data's size to 12×12. Second convolution, multiplies the size of feature space to 6 and their sizes to 8×8 with pooling reducing it further to 4×4. The number of feature maps are independently characterized and decides the convergence rate of the model alongside the learning rate.

Confusion Matrix

Confusion Matrix have prediction and actual values as the independent and dependent axes respectively. A diagonal layout of frequency maxima means they're predicting the "expected" output, as seen in the right plot of 1